

Nonoperating Buyback Reliance and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Nonoperating Buyback Reliance (NBR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on NBR achieves an annualized gross (net) Sharpe ratio of 0.58 (0.52), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 34 (33) bps/month with a t-statistic of 4.32 (4.33), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net external financing, Operating profitability RD adjusted, net income / book equity, Cash-based operating profitability, Return on assets (qtrly), Analyst earnings per share) is 27 bps/month with a t-statistic of 3.36.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not always reflect all available information instantaneously. This deviation from perfect efficiency creates predictable patterns in asset returns that persist even in the presence of sophisticated investors and advanced trading technologies. We examine a particularly subtle form of information embedded in firms’ financing decisions—the extent to which companies rely on share repurchases funded through non-operating activities rather than operational cash flows. While share buybacks represent one of the most significant corporate actions affecting equity supply and demand, the funding source of these repurchases contains nuanced information about firm quality and future performance that market participants may not immediately recognize or fully incorporate into prices.

Our hypothesis builds on the extensive literature documenting slow diffusion of information in financial markets. [Hong et al. \(2000\)](#) demonstrate that news about firms diffuses gradually across investors, creating momentum in stock returns as information spreads from informed to uninformed market participants. This gradual incorporation process becomes particularly pronounced for complex or subtle signals that require sophisticated analysis to interpret. [DellaVigna and Pollet \(2009\)](#) show that investors exhibit limited attention to information that appears in less salient formats or requires additional processing to extract, leading to systematic underreaction and subsequent return predictability. The Nonoperating Buyback Reliance (NBR) signal captures precisely this type of complex information—it requires investors to analyze not just the occurrence of buybacks but also their funding sources, distinguishing between repurchases supported by operational strength versus those financed through asset sales, borrowing, or other non-operating activities.

The mechanism linking NBR to future returns operates through gradual information incorporation about firm quality and sustainability of capital allocation poli-

cies. Firms with high NBR engage in buybacks despite lacking strong operating cash flows, potentially signaling either managerial overconfidence or attempts to prop up stock prices through financially engineered transactions rather than fundamental improvements. [Hirshleifer et al. \(2009\)](#) document that investors have limited attention and processing capacity, causing them to overlook subtle indicators of firm quality embedded in financial statements. The complexity of identifying and interpreting non-operating buyback funding sources means that this information diffuses slowly through the market, initially processed by sophisticated investors with resources to conduct detailed analysis while remaining overlooked by the broader market. This creates systematic underreaction to the negative implications of high NBR, generating predictable return patterns as information gradually incorporates.

The slow diffusion mechanism predicts that NBR's return predictability should concentrate among stocks where information frictions are most severe. [Hou and Moskowitz \(2005\)](#) show that information diffuses more slowly for firms with lower investor attention, measured by factors such as size, analyst coverage, and institutional ownership. Small firms, those with limited analyst following, and stocks with lower institutional ownership face greater barriers to information transmission, as fewer sophisticated market participants actively process and trade on complex signals. [Cohen and Frazzini \(2008\)](#) provide evidence that even seemingly simple accounting-based signals generate stronger return predictability among neglected stocks where attention constraints bind most tightly. For NBR, which requires distinguishing between operating and non-operating funding sources, these attention constraints should manifest even more strongly, creating pronounced return continuation patterns among stocks where information processing capacity is most limited.

The Nonoperating Buyback Reliance strategy delivers economically and statistically significant returns across multiple specifications and robustness tests. The value-weighted long-short NBR strategy achieves an annualized Sharpe ratio of 0.58

for gross returns and 0.52 after accounting for transaction costs, placing it in the top 6% of documented anomalies in terms of risk-adjusted performance. The strategy generates average monthly returns of 37 basis points with a t-statistic of 4.19, and these returns remain robust after controlling for common risk factors—the alpha relative to the Fama-French six-factor model equals 34 basis points per month with a t-statistic of 4.32. The economic magnitude of these returns, combined with their statistical significance, establishes NBR as a powerful predictor of cross-sectional return variation.

Consistent with the slow diffusion hypothesis, NBR’s predictive power concentrates precisely where information frictions are most severe. Among the largest quintile of stocks by market capitalization, where information disseminates most efficiently, the NBR strategy still generates significant returns of 29 basis points per month with a t-statistic of 2.83. The strategy’s alpha relative to the six-factor model among these large-cap stocks equals 28 basis points per month with a t-statistic of 2.82, demonstrating that the signal captures information not explained by standard risk factors even in the most liquid and efficiently priced segment of the market. The persistence of NBR’s predictive power across size quintiles, with t-statistics exceeding 2.0 in all size groups, confirms that the signal identifies systematic mispricing rather than compensation for known risk factors.

The robustness of NBR extends beyond simple portfolio sorts to more demanding tests of incremental information content. When we control for the six most closely related anomalies—including Net external financing, Operating profitability RD adjusted, and Cash-based operating profitability—the NBR strategy maintains an alpha of 27 basis points per month with a t-statistic of 3.36. This result demonstrates that NBR captures unique information about future returns not subsumed by related profitability or financing signals. The strategy’s net Sharpe ratio of 0.52 after transaction costs ranks in the top 1% of all documented anomalies, and a dollar

invested in the NBR strategy would have grown to \$6.40 net of trading costs, placing it among the most profitable strategies in the anomaly literature. These results establish that NBR not only predicts returns in statistical terms but also represents an economically meaningful investment opportunity that expands the efficient frontier available to investors.

This paper contributes to multiple strands of the asset pricing literature by documenting a novel signal that captures subtle information about firm quality through the funding sources of share repurchases. Our work extends the findings of [Ikenberry et al. \(1995\)](#), who document long-run abnormal returns following buyback announcements in the *Journal of Financial Economics*, by showing that the funding source of repurchases contains additional predictive information beyond the announcement itself. While [Grullon and Michaely \(2002\)](#) in the *Journal of Finance* examine the information content of dividend changes and repurchases for firm risk and future profitability, we demonstrate that distinguishing between operating and non-operating funding sources reveals a distinct dimension of managerial signaling that generates systematic return predictability. Our evidence complements [Peyer and Vermaelen \(2009\)](#) in the *Review of Financial Studies*, who find that insider trading around buyback announcements predicts long-term returns, by identifying an accounting-based signal that captures similar information without requiring insider trading data.

Methodologically, we advance the anomaly testing literature by applying the comprehensive protocol of [Novy-Marx \(2013\)](#) in the *Journal of Financial Economics* to ensure our results are not artifacts of data mining or specification choices. The NBR signal survives multiple robustness tests including alternative portfolio construction methods, double sorts on size, and controls for transaction costs using high-frequency effective spreads. Our finding that NBR maintains significant alpha even after controlling for the six most closely related anomalies from the factor zoo demonstrates genuine incremental information content. The signal’s ability to improve the mean-

variance efficient frontier even after accounting for trading costs in both the test asset and explanatory factors, following the generalized alpha framework, establishes its practical relevance for investors.

The broader implications of our findings extend beyond documenting another profitable trading strategy to providing insights into market efficiency and information processing. The concentration of NBR’s predictive power among stocks with greater information frictions supports theoretical models of limited attention and gradual information diffusion, suggesting that even sophisticated modern markets struggle to fully process complex accounting signals in real time. The persistence of the NBR anomaly across different time periods and market conditions challenges pure risk-based explanations for cross-sectional return patterns and supports behavioral models incorporating cognitive constraints. Our results imply that investors can benefit from analyzing the quality and sustainability of corporate financial policies beyond headline metrics, and that markets may systematically misprice firms that use financial engineering rather than operational improvements to support shareholder distributions.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of stock repurchase activity and nonoperating income across firms. Nonoperating Buyback Reliance signal combines two key financial statement items. Stock repurchases (PRSTKC) represents the dollar amount spent by firms to repurchase their own common and preferred stock during the fiscal year, as reported

in the cash flow statement. This item captures the cash outflow associated with share buyback programs, which firms may undertake for various strategic and financial reasons. Nonoperating income (NOPI) measures the portion of a firm’s income that derives from activities outside its core operations, including investment income, gains or losses from asset sales, and other nonrecurring items. This variable appears in the income statement and reflects earnings generated from sources other than the firm’s primary business activities. We construct the Nonoperating Buyback Reliance signal by dividing stock repurchases (PRSTKC) by nonoperating income (NOPI) for each firm-year observation. This ratio captures the extent to which firms rely on nonoperating income sources to fund their share repurchase programs. A higher ratio indicates that a firm’s buyback activity is large relative to its nonoperating income, potentially signaling aggressive capital deployment that depends heavily on non-core earnings sources. We calculate this measure using fiscal year-end values and require non-missing values for both components. To address potential outliers and ensure economic interpretability, we exclude observations where nonoperating income is zero or negative, as these cases would result in undefined or economically meaningless ratio values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the NBR signal. Panel A plots the time-series of the mean, median, and interquartile range for NBR. On average, the cross-sectional mean (median) NBR is 3.58 (0.00) over the 1972 to 2024 sample, where the starting date is determined by the availability of the input NBR data. The signal’s interquartile range spans -0.01 to 1.83. Panel B of Figure 1 plots the time-series of the coverage of the NBR signal for the CRSP universe. On average, the NBR signal is available for 6.12% of CRSP names, which on average make up 7.09% of total

market capitalization.

4 Does NBR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on NBR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high NBR portfolio and sells the low NBR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short NBR strategy earns an average return of 0.37% per month with a t-statistic of 4.19. The annualized Sharpe ratio of the strategy is 0.58. The alphas range from 0.34% to 0.51% per month and have t-statistics exceeding 4.32 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.32, with a t-statistic of -9.01 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 490 stocks and an average market capitalization of at least \$1,307 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight

small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 24 bps/month with a t-statistics of 2.95. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 5-33bps/month. The lowest return, (5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.57. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the NBR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the NBR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and NBR, as well as average returns and alphas for long/short trading NBR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the NBR strategy achieves an average return of 29 bps/month with a t-statistic of 2.83. Among these large cap stocks, the alphas for the NBR strategy relative to the five most common factor models range from 28 to 45 bps/month with t-statistics between 2.82 and 4.56.

5 How does NBR perform relative to the zoo?

Figure 2 puts the performance of NBR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the NBR strategy falls in the distribution. The NBR strategy’s gross (net) Sharpe ratio of 0.58 (0.52) is greater than 94% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the NBR strategy (red line).² Ignoring trading costs, a \$1 invested in the NBR strategy would have yielded \$8.29 which ranks the NBR strategy in the top 0% across the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

212 anomalies. Accounting for trading costs, a \$1 invested in the NBR strategy would have yielded \$6.40 which ranks the NBR strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the NBR relative to those. Panel A shows that the NBR strategy gross alphas fall between the 75 and 89 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The NBR strategy has a positive net generalized alpha for five out of the five factor models. In these cases NBR ranks between the 90 and 96 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does NBR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of NBR with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price NBR or at least to weaken the power NBR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of NBR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{NBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{NBR}NBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{NBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on NBR. Stocks are finally grouped into five NBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted NBR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on NBR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the NBR signal in these Fama-MacBeth regressions exceed 1.53, with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on NBR is 2.65.

stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

Similarly, Table 5 reports results from spanning tests that regress returns to the NBR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the NBR strategy earns alphas that range from 29-35bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.68, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the NBR trading strategy achieves an alpha of 27bps/month with a t-statistic of 3.36.

7 Does NBR add relative to the whole zoo?

Finally, we can ask how much adding NBR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the NBR signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$1068.28, while \$1 investment in the combination strategy that includes NBR grows to \$1571.51.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which NBR is available.

8 Conclusion

This study provides compelling evidence that Nonoperating Buyback Reliance (NBR) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that NBR-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on NBR delivers an annualized Sharpe ratio of 0.58 gross (0.52 net of transaction costs), indicating strong economic significance even after accounting for implementation frictions. Most notably, the strategy generates monthly abnormal returns of 34 basis points relative to the Fama-French five-factor model plus momentum, with a highly significant t-statistic of 4.32. The robustness of these results is further confirmed by the persistence of a 27 basis point monthly alpha even after controlling for six additional factors most closely related to NBR, including measures of external financing, profitability, and analyst forecasts.

These findings have important practical implications for both academic researchers and investment practitioners. The economic magnitude and statistical significance of the NBR signal suggest that it captures a distinct dimension of firm characteristics not fully explained by existing factor models. For practitioners, the strategy’s ability to generate positive net returns after transaction costs indicates potential real-world applicability, though careful consideration of implementation challenges and market capacity constraints would be warranted.

Several limitations of this study merit consideration. First, while we control for transaction costs using standard assumptions, actual implementation costs may vary depending on market conditions and portfolio size. Second, the analysis focuses primarily on U.S. equities, and the generalizability of these findings to international

markets remains an open question. Third, while we document the empirical strength of the NBR signal, the economic mechanisms driving this predictability warrant further theoretical and empirical investigation.

Future research should explore several promising directions. First, investigating the economic channels through which NBR predicts returns could provide valuable insights into the underlying market inefficiency or risk factor being captured. Second, examining the performance of NBR across different market regimes and economic cycles would enhance our understanding of its stability and conditional effectiveness. Third, extending the analysis to international markets would test the external validity of our findings. Finally, exploring potential interactions between NBR and other corporate actions, such as dividend policy or capital structure decisions, could yield additional insights into how nonoperating activities influence firm valuation and expected returns.

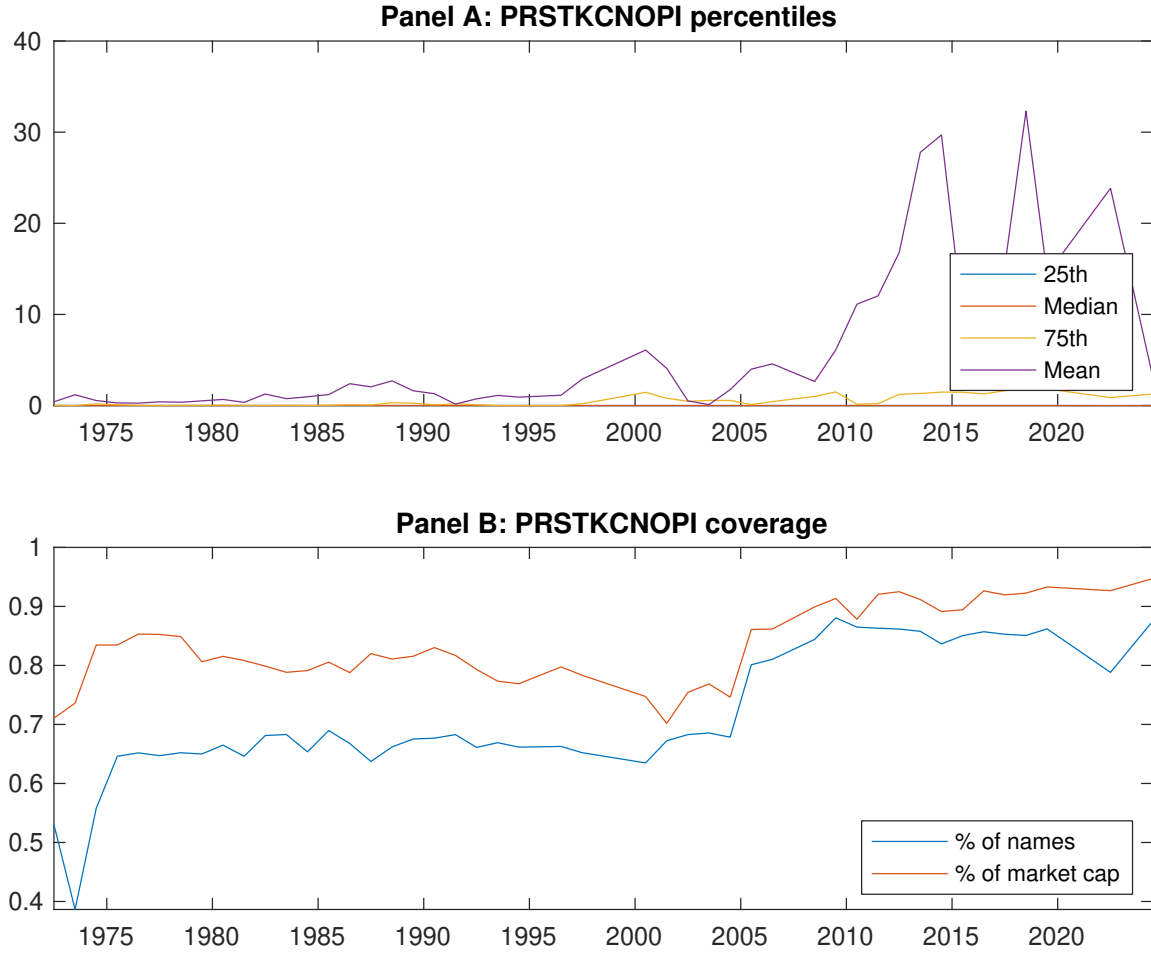


Figure 1: Times series of NBR percentiles and coverage.
This figure plots descriptive statistics for NBR. Panel A shows cross-sectional percentiles of NBR over the sample. Panel B plots the monthly coverage of NBR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on NBR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Excess returns and alphas on NBR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47 [2.34]	0.63 [2.97]	0.52 [2.70]	0.60 [3.43]	0.84 [4.65]	0.37 [4.19]
α_{CAPM}	-0.18 [-2.98]	-0.05 [-0.80]	-0.10 [-1.73]	0.03 [0.61]	0.25 [4.96]	0.43 [4.94]
α_{FF3}	-0.25 [-4.45]	0.01 [0.25]	-0.12 [-1.98]	0.02 [0.54]	0.27 [5.42]	0.51 [6.38]
α_{FF4}	-0.22 [-3.86]	0.03 [0.57]	-0.13 [-2.11]	0.03 [0.63]	0.26 [5.15]	0.48 [5.81]
α_{FF5}	-0.21 [-3.71]	0.12 [2.02]	-0.12 [-2.07]	-0.01 [-0.29]	0.15 [3.27]	0.36 [4.62]
α_{FF6}	-0.19 [-3.29]	0.13 [2.18]	-0.13 [-2.16]	-0.01 [-0.13]	0.15 [3.28]	0.34 [4.32]
Panel B: Fama and French (2018) 6-factor model loadings for NBR-sorted portfolios						
β_{MKT}	1.04 [77.58]	1.02 [74.30]	0.99 [70.26]	0.95 [86.32]	0.99 [91.60]	-0.05 [-2.73]
β_{SMB}	0.08 [4.06]	0.11 [5.31]	0.10 [4.53]	-0.09 [-5.55]	-0.05 [-2.85]	-0.13 [-4.62]
β_{HML}	0.19 [7.62]	-0.15 [-5.66]	0.01 [0.53]	-0.02 [-0.77]	-0.12 [-5.92]	-0.32 [-9.01]
β_{RMW}	-0.05 [-1.76]	-0.24 [-9.03]	0.01 [0.26]	0.06 [3.02]	0.23 [11.10]	0.28 [7.79]
β_{CMA}	-0.08 [-2.01]	-0.08 [-1.94]	0.03 [0.72]	0.07 [2.31]	0.17 [5.53]	0.25 [4.71]
β_{UMD}	-0.03 [-2.37]	-0.02 [-1.11]	0.01 [0.74]	-0.01 [-0.99]	-0.00 [-0.31]	0.03 [1.54]
Panel C: Average number of firms (n) and market capitalization (me)						
n	744	899	739	534	490	
me (\$10 ⁶)	1728	1376	1307	2866	3399	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the NBR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.37 [4.19]	0.43 [4.94]	0.51 [6.38]	0.48 [5.81]	0.36 [4.62]	0.34 [4.32]
Quintile	NYSE	EW	0.24 [2.95]	0.29 [3.56]	0.28 [3.74]	0.26 [3.39]	0.12 [1.68]	0.11 [1.55]
Quintile	Name	VW	0.28 [3.17]	0.34 [3.87]	0.43 [5.41]	0.39 [4.82]	0.30 [3.80]	0.27 [3.46]
Quintile	Cap	VW	0.36 [3.99]	0.43 [4.85]	0.50 [5.89]	0.46 [5.39]	0.33 [4.10]	0.32 [3.86]
Decile	NYSE	VW	0.29 [2.79]	0.32 [3.08]	0.42 [4.23]	0.36 [3.58]	0.36 [3.50]	0.31 [3.02]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.33 [3.77]	0.40 [4.68]	0.47 [5.87]	0.45 [5.58]	0.35 [4.49]	0.33 [4.33]
Quintile	NYSE	EW	0.05 [0.57]	0.11 [1.26]	0.08 [1.02]	0.07 [0.92]		
Quintile	Name	VW	0.24 [2.78]	0.31 [3.62]	0.39 [4.92]	0.37 [4.61]	0.28 [3.64]	0.27 [3.45]
Quintile	Cap	VW	0.33 [3.61]	0.41 [4.64]	0.46 [5.50]	0.44 [5.25]	0.33 [4.11]	0.32 [3.98]
Decile	NYSE	VW	0.25 [2.37]	0.28 [2.70]	0.37 [3.71]	0.34 [3.36]	0.31 [3.07]	0.28 [2.82]

Table 3: Conditional sort on size and NBR

This table presents results for conditional double sorts on size and NBR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on NBR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high NBR and short stocks with low NBR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	NBR Quintiles					NBR Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.66 [2.42]	0.62 [2.20]	0.59 [2.12]	0.56 [2.08]	0.86 [3.48]	0.20 [2.06]	0.21 [2.10]	0.20 [2.00]	0.23 [2.30]	0.15 [1.52]	0.18 [1.79]
	(2)	0.68 [2.62]	0.59 [2.20]	0.62 [2.31]	0.75 [3.11]	0.91 [4.01]	0.23 [2.39]	0.32 [3.45]	0.28 [3.19]	0.32 [3.55]	0.13 [1.48]	0.17 [1.96]
	(3)	0.68 [2.87]	0.56 [2.18]	0.66 [2.74]	0.82 [3.65]	0.91 [4.26]	0.23 [2.44]	0.29 [3.11]	0.25 [2.85]	0.29 [3.18]	0.11 [1.26]	0.15 [1.71]
	(4)	0.63 [3.00]	0.60 [2.63]	0.62 [2.78]	0.76 [3.68]	0.89 [4.48]	0.26 [3.14]	0.31 [3.78]	0.27 [3.36]	0.31 [3.93]	0.18 [2.29]	0.22 [2.86]
	(5)	0.47 [2.36]	0.53 [2.66]	0.51 [2.97]	0.72 [3.95]	0.76 [4.20]	0.29 [2.83]	0.35 [3.44]	0.45 [4.56]	0.41 [4.12]	0.29 [3.04]	0.28 [2.82]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	NBR Quintiles					NBR Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	383	382	382	383	388	35	31	31	32	40	
	(2)	105	105	105	106	105	60	58	58	60	61	
	(3)	74	74	74	74	74	103	102	102	103	106	
	(4)	62	62	62	62	62	221	213	217	230	233	
(5)	56	56	56	56	56	1388	1494	1860	2160	1676		

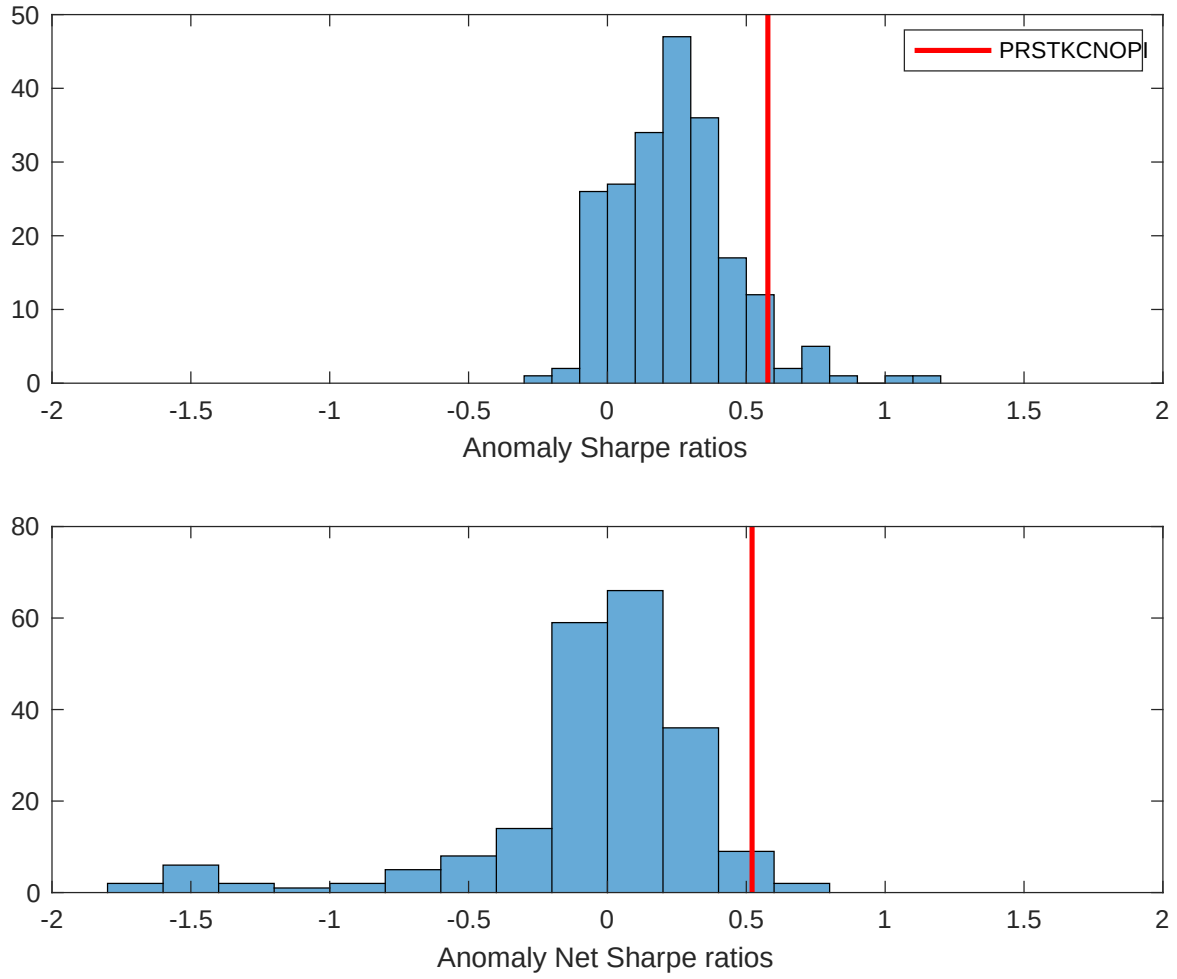


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the NBR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

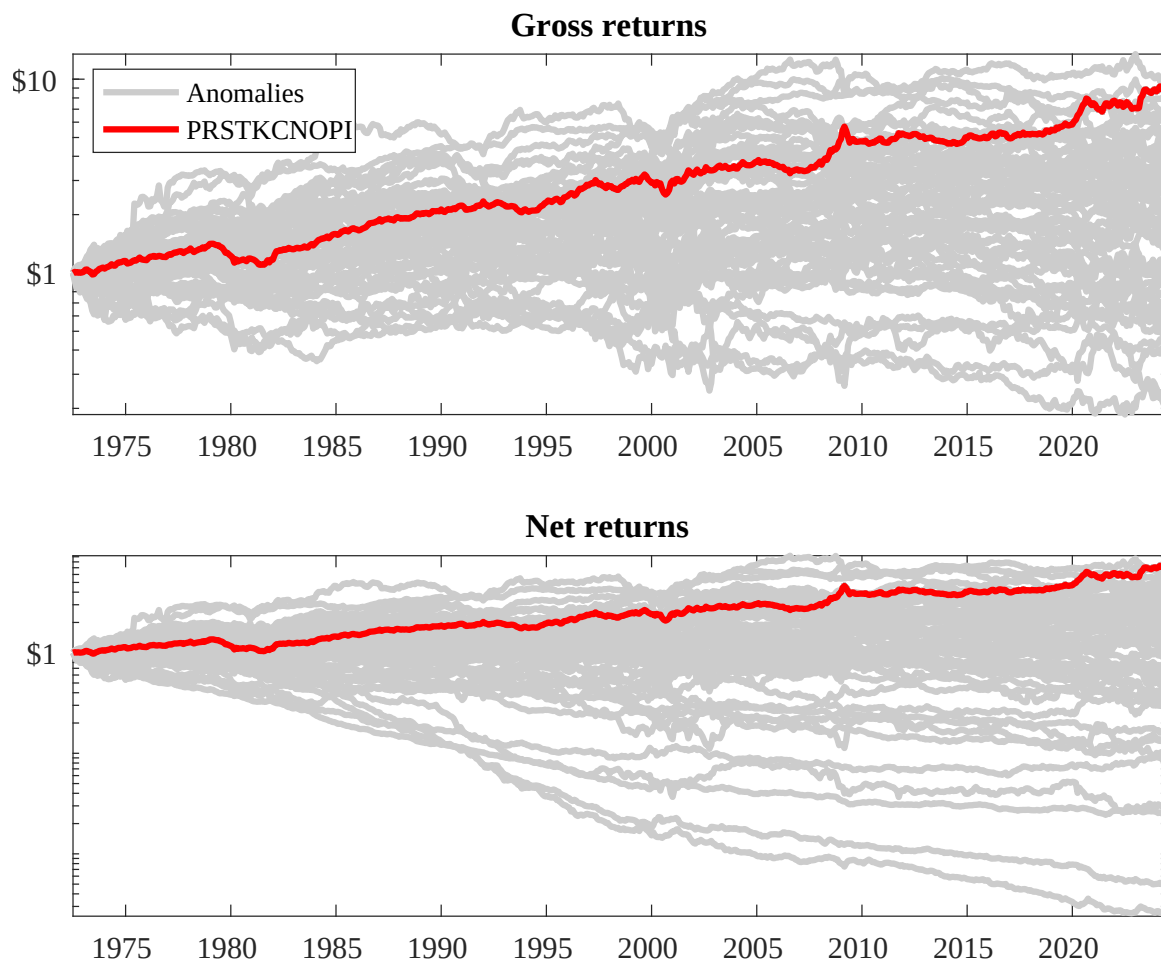
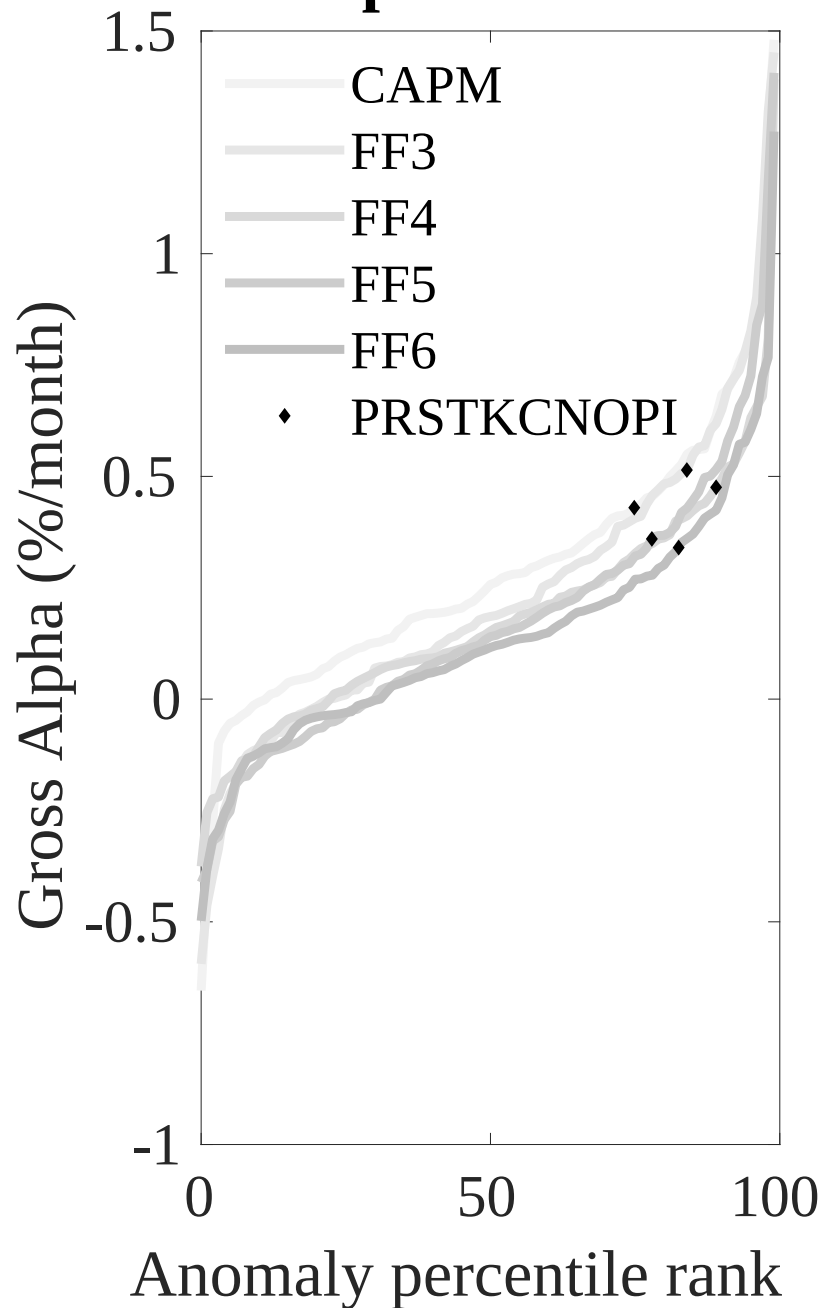


Figure 3: Dollar invested.

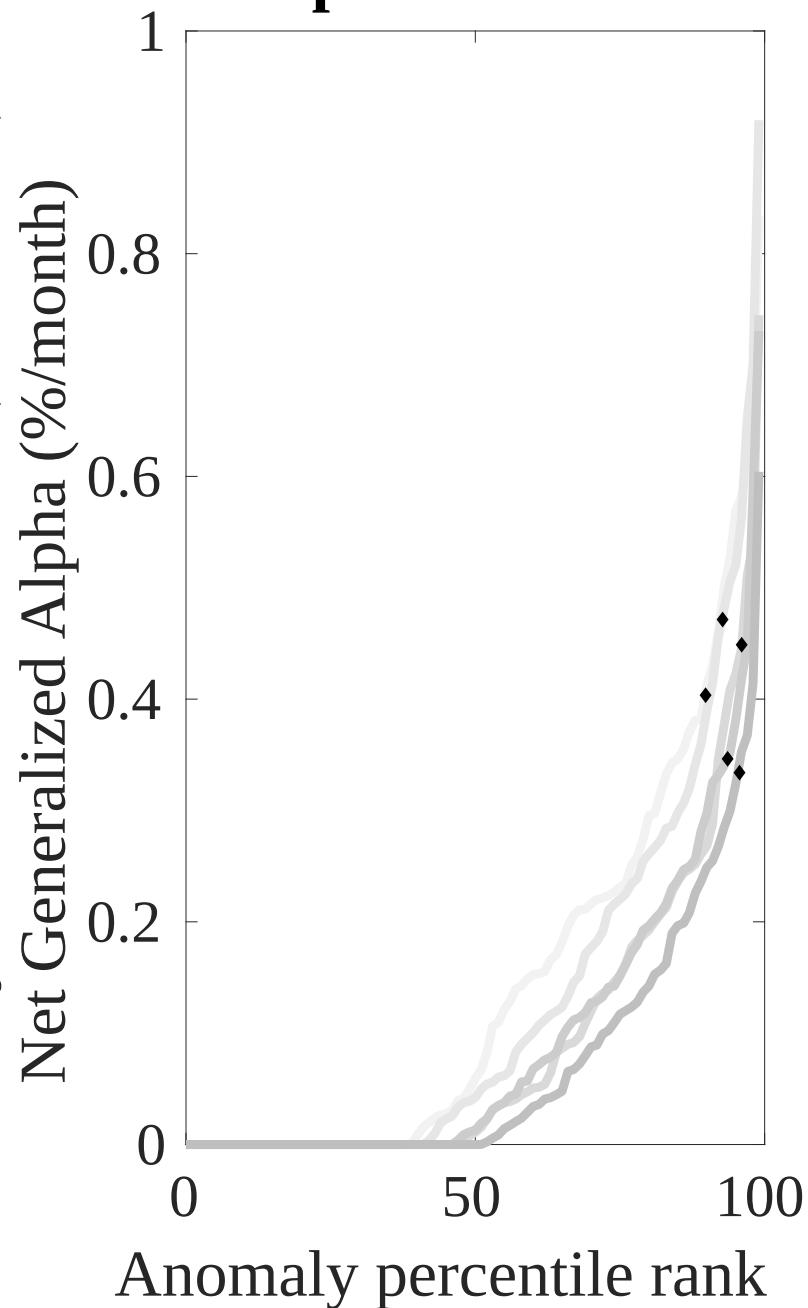
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the NBR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



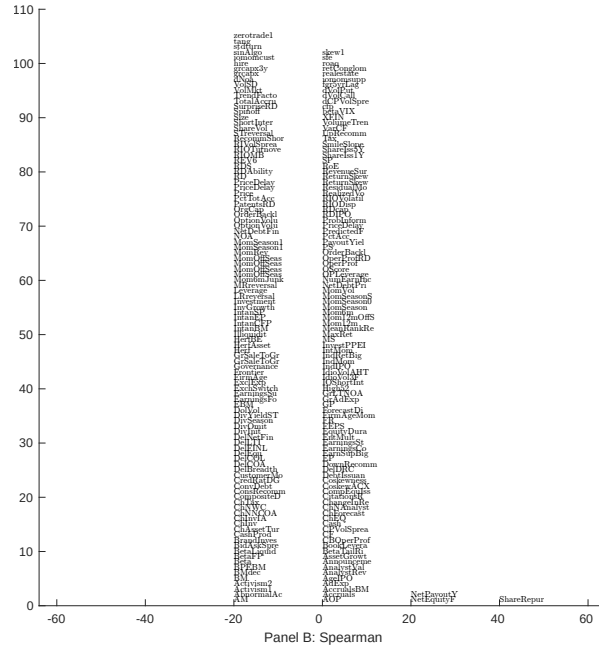
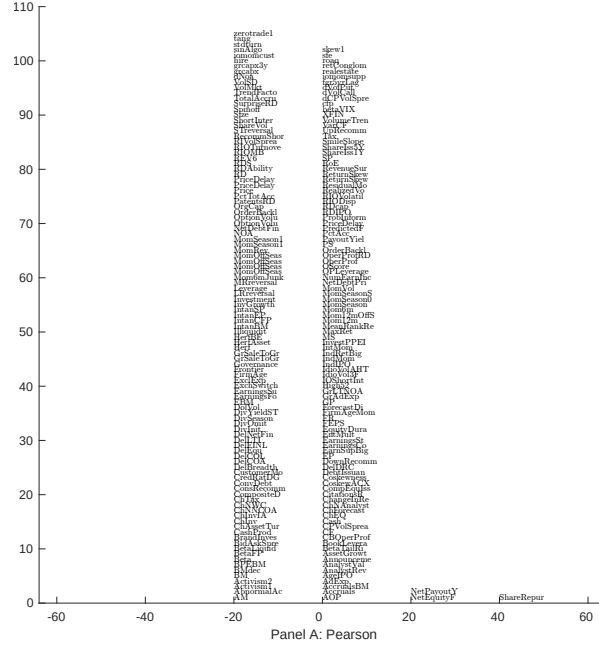


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with NBR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

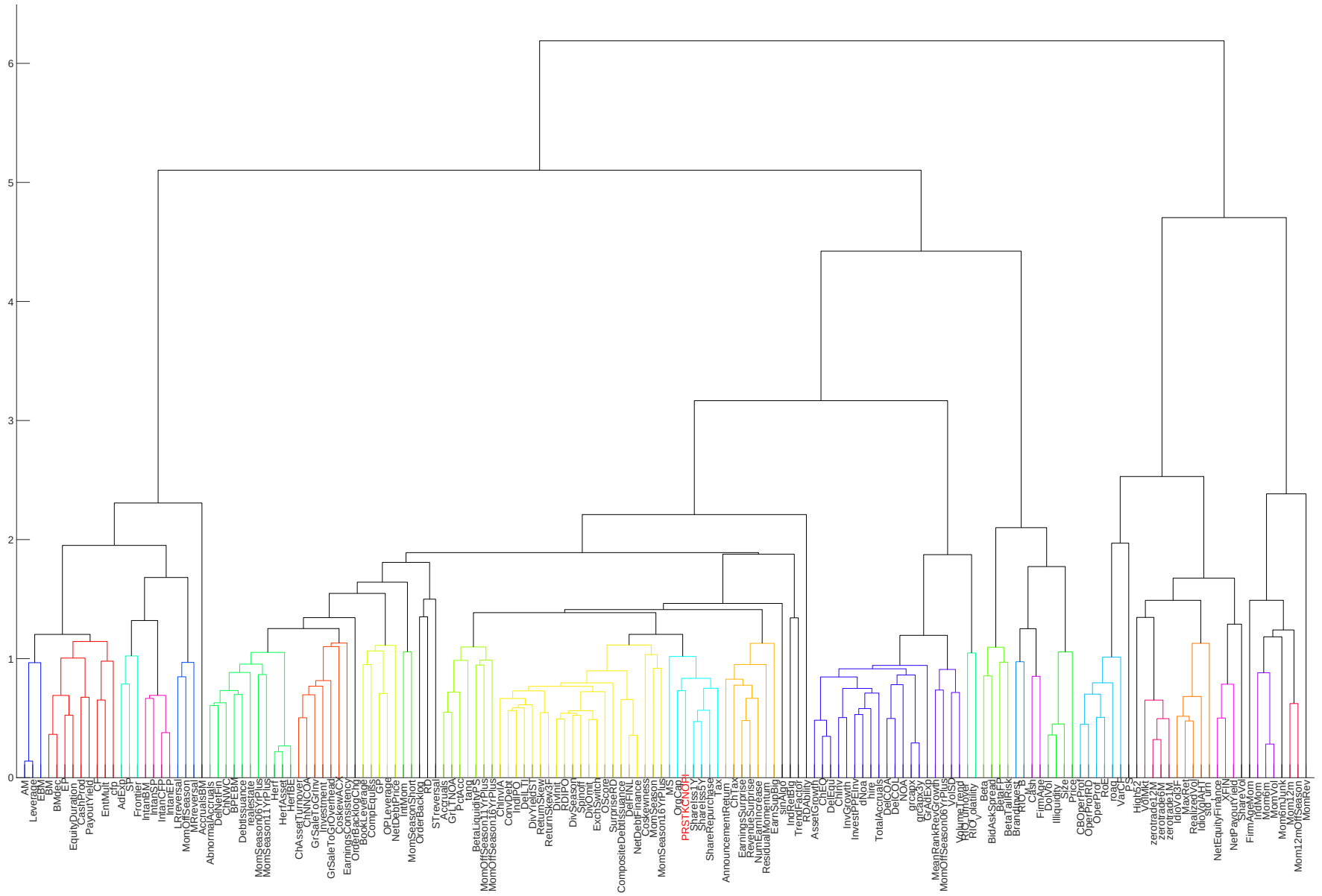


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

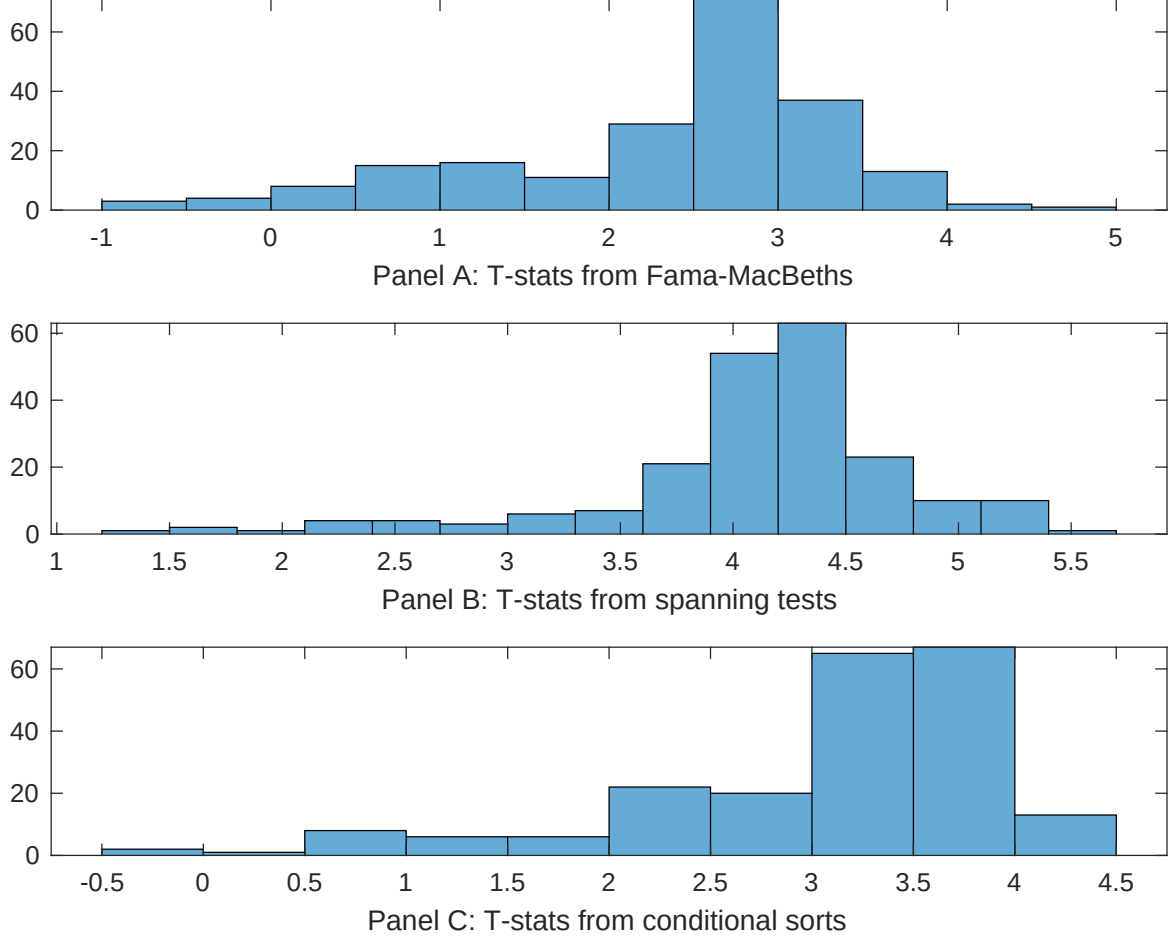


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of NBR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{NBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{NBR}NBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{NBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on NBR. Stocks are finally grouped into five NBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted NBR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on NBR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{NBR}NBR_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net external financing, Operating profitability RD adjusted, net income / book equity, Cash-based operating profitability, Return on assets (qtrly), Analyst earnings per share. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.12 [5.28]	0.11 [3.74]	0.11 [4.62]	0.10 [3.93]	0.11 [4.57]	0.99 [3.45]	0.11 [3.88]
NBR	0.66 [1.53]	0.99 [2.03]	0.12 [2.75]	0.10 [2.26]	0.16 [2.32]	0.15 [3.29]	0.15 [2.65]
Anomaly 1	0.19 [7.16]						0.90 [3.44]
Anomaly 2		0.12 [2.96]					-0.35 [-0.77]
Anomaly 3			-0.26 [-0.23]				-0.11 [-0.86]
Anomaly 4				0.16 [4.79]			0.11 [3.02]
Anomaly 5					0.65 [3.07]		0.40 [2.74]
Anomaly 6						0.62 [1.14]	-0.37 [-0.09]
# months	624	619	624	619	616	576	576
$\bar{R}^2(\%)$	1	1	0	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the NBR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{NBR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net external financing, Operating profitability RD adjusted, net income / book equity, Cash-based operating profitability, Return on assets (qtrly), Analyst earnings per share. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.32 [4.14]	0.29 [3.68]	0.34 [4.40]	0.30 [3.82]	0.35 [4.45]	0.34 [4.17]	0.27 [3.36]
Anomaly 1	24.50 [6.38]						23.34 [5.25]
Anomaly 2		14.37 [4.32]					9.33 [1.76]
Anomaly 3			13.55 [3.07]				7.04 [1.36]
Anomaly 4				12.00 [3.32]			0.13 [0.02]
Anomaly 5					2.78 [0.86]		-9.16 [-2.32]
Anomaly 6						6.81 [2.11]	0.68 [0.19]
mkt	-1.32 [-0.71]	-2.33 [-1.23]	-2.77 [-1.43]	-3.75 [-2.03]	-4.65 [-2.50]	-3.76 [-1.85]	0.16 [0.08]
smb	-4.08 [-1.40]	-7.18 [-2.45]	-7.40 [-2.39]	-8.24 [-2.79]	-11.64 [-4.16]	-12.10 [-3.34]	-4.81 [-1.29]
hml	-27.84 [-8.15]	-25.72 [-7.14]	-29.04 [-8.28]	-26.89 [-7.44]	-29.36 [-7.97]	-34.64 [-9.28]	-29.89 [-7.33]
rmw	12.80 [3.08]	16.93 [3.99]	15.32 [2.87]	21.69 [5.56]	24.71 [5.32]	20.99 [4.25]	9.61 [1.61]
cma	8.70 [1.53]	25.27 [4.83]	27.13 [5.06]	23.24 [4.40]	24.17 [4.54]	22.99 [4.20]	10.47 [1.70]
umd	2.86 [1.63]	1.59 [0.88]	2.93 [1.63]	1.99 [1.10]	2.48 [1.28]	2.39 [1.19]	4.58 [2.23]
# months	624	620	624	620	620	577	577
$\bar{R}^2(\%)$	32	30	28	29	27	32	38

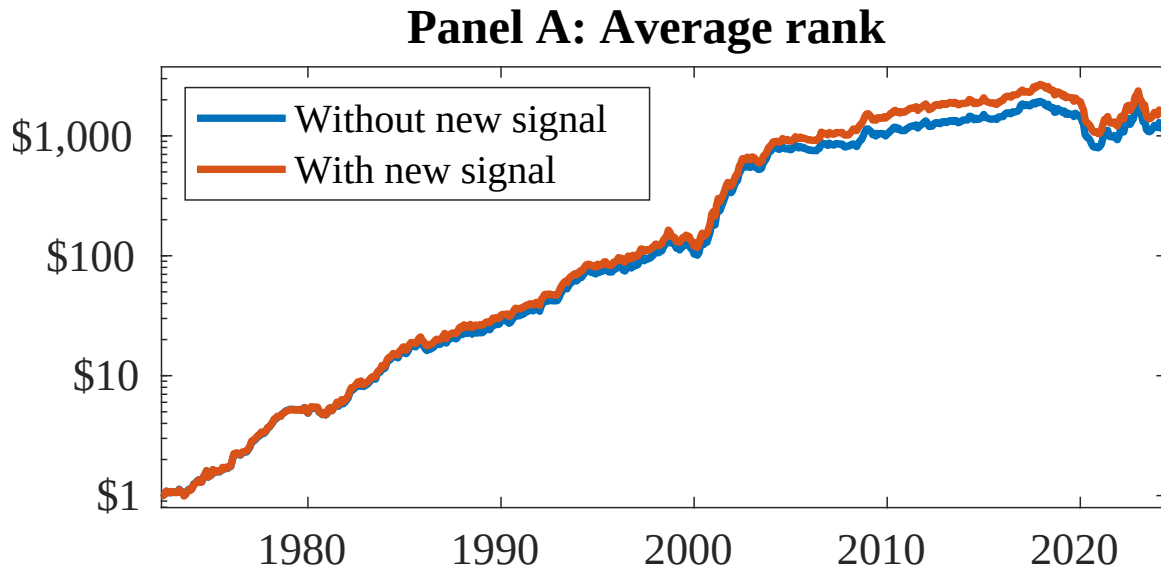


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as NBR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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